

INDIAN-EQUITY QUANT RESEARCH & PAPER-TRADING PLATFORM

# MICC

From a 19.2 GB market-data warehouse to a walk-forward-validated, paper-traded, self-scoring strategy platform.

88 tables · 146.7M rows · survivorship-free since 2005 · research only, not advice

## **The one-line thesis**

Clean data → no-lookahead features → 8 strategies  
→ cost- & regime-aware backtest → forward paper-  
trade → trackable recommendations → every claim  
verified by a test.

One proven edge (momentum, 20 yr). Everything else is held to the same honest bar —  
including the failures.

## 1 • The data

| Layer               | Highlights                                                                                    |
|---------------------|-----------------------------------------------------------------------------------------------|
| Equity              | 7.65M rows · 4,200 symbols · 2005→2026 · <b>survivorship-free</b> (delisted kept) · delivery% |
| Derivatives         | 68.9M-row F&O bhavcopy · PCR · participant OI                                                 |
| Mutual funds        | 36.9M NAVs · 37,977 schemes                                                                   |
| Macro / cross-asset | US from 1919 · IndiaVIX, NIFTY, S&P, USDINR, Brent, US10Y                                     |
| Events              | insider (SEBI) · bulk/block deals · corporate actions                                         |

Binding limit: fundamentals only from 2021 (shallow).

## 2 · Cleaning — turning raw into trustworthy

- ▶ **Adjusted prices** (split/bonus) with a **cliff-verification guard** — only adjusts when the raw price actually shows the drop, so it never manufactures a fake cliff on already-adjusted names.
- ▶ **Total-return** series (dividends): validated COALINDIA +7.6%/yr.
- ▶ **Point-in-time universe**: survivorship-free, ETFs excluded (ISIN INF vs INE).
- ▶ **PIT fundamentals**: every figure tagged with its filing date → no lookahead.

## 3 • Features — predictive & leak-free

As-of cross-sectional rank-IC vs forward return (top-500, 257 months):

| Feature                | IC (1m) | IC (3m) |
|------------------------|---------|---------|
| 52-week-high proximity | +0.051  | +0.090  |
| Low volatility         | +0.058  | —       |
| Delivery %             | +0.045  | +0.065  |
| 12-1 momentum          | +0.041  | +0.052  |
| Realized vol           | −0.051  | −0.072  |

Every sign matches theory; magnitudes  $\sim 0.05$  = legitimate, not lookahead-inflated.

## 4 • The 8-strategy library

| Strategy                 | Sharpe | CAGR  | MaxDD |
|--------------------------|--------|-------|-------|
| value_earnings_yield     | 1.57   | 28.4% | -14%  |
| momentum_delivery_lowvol | 1.46   | 17.6% | -16%  |
| quality_roe              | 1.30   | 18.7% | -9%   |
| dividend_yield           | 0.92   | 15.1% | -39%  |
| sector_rotation          | 0.87   | 13.0% | -45%  |
| low_volatility           | 0.83   | 8.3%  | -21%  |
| high52_breakout          | 0.82   | 11.0% | -31%  |
| short_term_reversal      | -0.12  | -4.9% | -87%  |

⚠ value/quality have only ~3–4 yr data. **Momentum is the one proven 20-yr edge.**

## | 5 • Timing — the macro regime gate

4 risk-on votes:  $\text{breadth} > 200\text{DMA} \cdot \text{NIFTY} > 200\text{DMA} \cdot \text{S\&P} > 200\text{DMA} \cdot \text{India VIX low}$ .

Invest when  $\geq 2$  votes, else hold cash. Walk-forward-validated: regime gate

**OOS Sharpe 1.56** vs breadth-only 1.34. **Live now: 1/4 → RISK-OFF.**

## 6 • Validation — the honest bar

- ▶ Purged + embargoed **walk-forward** (forward label can't leak)
- ▶ Stable in every sub-period (2005–11, 12–18, 19–26)
- ▶ **Deflated Sharpe**  $\approx$  **100%** after multiple-testing — not data-mined
- ▶ Capacity (sqrt-impact)  $\approx$  **₹100–250 cr**
- ▶ **ML ranker LOST** to the simple model (OOS 0.76 vs 1.25) — research-first



## 7 • The validated best config

**OOS Sharpe 1.53**

inverse-vol weighting + walk-forward regime gate · Calmar 1.26 · MaxDD  
– 18.9% · 41.6× equity (2009→2026)

## 8 · Paper trading — forward, with frictions

**₹10L → ₹3.2 cr**

integer shares · real Indian costs · regime liquidation to cash · drift vs backtest  
= 1.000

At today's RISK-OFF regime, the paper book is sitting in all cash.

## 9 • Recommendations + feedback loop

Each call = entry / target / stop price band + a 1-month duration → scored against the real price path  
→ a track record that improves the model.

| Closed calls | Hit rate | Avg / call |
|--------------|----------|------------|
| 525          | 49%      | +0.47%     |

Insight it already surfaced: tight vol-stops whipsaw the momentum edge (34% stop-hit).

## 10 · Execution & safety

- ▶ **OMS + RiskEngine**: rejects oversized ( $>6\%$ ), illiquid ( $>10\%$  ADV), and dust orders.
- ▶ **PaperBroker** simulates fills.
- ▶ **LiveBroker is DISABLED** — refuses any real order without your API keys + an explicit enable. No accidental real money.

## 11 • Delivery

- ▶ Self-contained **HTML dashboard** (regime, leaderboard, equity curve, recommendations, funds, deals, F&O)
- ▶ Authenticated **web server** :8765 + JSON **REST API** /api/\*
- ▶ **run\_pipeline.py** — daily refresh + weekly full rebuild

## 12 · Verification — nothing trusted without a test

29 + 17 + 12

29 pin-to-pin · 17 deep integrity · 12 daily health checks — all green.

Including: value strategy 0/1,758 lookahead violations · risk engine proven to reject bad orders · paper↔backtest drift 1.000.

## 13 • The honest standing

| Verdict          | What                                                          |
|------------------|---------------------------------------------------------------|
| <b>PROVEN</b>    | Momentum + delivery + low-vol, regime-gated (20 yr, OOS 1.53) |
| <b>PROMISING</b> | Value / quality — high Sharpe but only 3–4 yr data            |
| <b>MODEST</b>    | Dividend, sector-rotation (deep drawdowns)                    |
| <b>DEAD</b>      | Short-term reversal, naive long-short (shown, not hidden)     |
| <b>GATED</b>     | Live trading — paper only until broker wired                  |

## Appendix A · Cliff-verification guard

Why adjusted prices are correct — adjust only when the raw price *really* dropped:

```
observed = close[on-ex] / close[ex-1]
apply factor iff |log(observed) - log(factor)| < |log(observed)|
```

DBEIL (9:1 bonus, factor 0.1) showed no raw cliff (observed 1.05) → closer to 1 → **skipped**, so it never manufactured a fake 10× drop. 1,130 applied · 41 skipped-already-adjusted.



## Appendix B · Regime walk-forward

The gate threshold is chosen from **past data only**, then applied forward:

```
for R at index i ≥ 48:  
    best_t = argmax over t∈{1..4} of Sharpe(gated returns on dates < R)  
    oos[R] = raw[R] if score[R] ≥ best_t else cash
```

Revealed the fixed  $\geq 2/4$  threshold is NOT robust (WF prefers  $\geq 1$ ) — yet the engine still beats breadth-only OOS 1.56 vs 1.34. A fixed-threshold backtest would have hidden that.

## Appendix C · Paper-trade drift fix

`paper.ret[R]` is the return realized **by** `R`; `backtest.net_gated[R]` is the **forward** return — off by one month:

```
bt["realized_by"] = bt["net_gated"].shift(1)  # = net_gated[R-1]
corr(paper.ret, bt.realized_by) → 0.24 becomes 1.000
```

A misaligned drift metric would falsely alarm. 1.000 proves the paper engine reproduces the backtest exactly.

## Appendix D • Engineering quality

- ▶ 88 tables · 146.7M rows · idempotent, re-runnable pipeline
- ▶ 7 real bugs caught & fixed mid-build ( $O(n^2)$  loop, fake-cliff, ETF leak, capacity calibration, ML merge bugs, drift off-by-one)
- ▶ Every claim has an executable test — 29 + 17 + 12 checks, all green
- ▶ Honest negatives kept, not hidden (failed strategies shown; ML loss documented)

# That's MICC.

Survivorship-free data · no-lookahead research · proven edge · forward-tested · self-scoring · fully verified.

Docs: [MICC\\_TECHNICAL\\_REPORT.md](#) · [MICC\\_BLUEPRINT.md](#) · [RESEARCH.md](#) · [README.md](#)